clear clk

rst_n

en

b_in_flat[31:0]

a_in_flat[31:0]

g_row[3].g_col[2].u_pe

pe_13

a_out_reg[0]_0

a_out_reg[2]_0

a_out_reg[2]_1

a_out_reg[3]_0
a_out_reg[3]_1

a_out_reg[4]_0

a_out_reg[4]_1

a_out_reg[5]_1

a_out_reg[5]_2

acc_reg[31]_0

clear clk

D[7:0] DI[2:0] Q[7:0] S[3:0] a_out_reg[1]_0[1:0] a_out_reg[1]_1[1:0] a_out_reg[2]_2[2:0] a_out_reg[2]_3[2:0] a_out_reg[5]_0[3:0] a_out_reg[6]_0

a_out_reg[6]_1

acc[15]_i_5__13[1:0] acc[15]_i_5__13_0[2:0] acc[3]_i_6__13_0[2:0] acc[3]_i_6__13_1[3:0] acc[3]_i_9__13_0[2:0] acc_reg[15]_0[3:0] acc_reg[19]_0[3:0] acc_reg[23]_0[3:0] acc_reg[27]_0[3:0] acc_reg[31]_1[3:0] b_out_reg[4] b_out_reg[4]_0[1:0] b_out_reg[7][2:0] b_out_reg[7]_0

c_out_flat[31:0] clear_0
product__59_carry_0[7:0] product__85_carry__0_i_3__13_0[1:0] product__85_carry__0_i_3__13_1[1:0] product__85_carry__0_i_4__13_0[3:0] product__85_carry__0_i_4__13_1[3:0] product__85_carry__0_i_6__13_0[3:0] product__85_carry__0_i_6__13_1[3:0] product__85_carry__1_i_2__13[1:0] product__85_carry__1_i_2__13_0[1:0] product__85_carry__1_i_3__13

product__85_carry__1_i_3__13_0

product__85_carry__1_i_5__13[3:0] product__85_carry_i_1__13_0[3:0] product__85_carry_i_8__13_0[3:0] product__85_carry_i_8__13_1[3:0]

g_row[3].g_col[3].u_pe

pe_14

acc_reg[0]_0

acc_reg[31]_0

clear clk

DI[2:0] S[3:0] acc[15]_i_5__14[1:0] acc[15]_i_5__14_0[2:0] acc[3]_i_6__14_0[2:0] acc[3]_i_6__14_1[3:0] acc[3]_i_9__14_0[2:0] acc_reg[15]_0[3:0] acc_reg[19]_0[3:0] acc_reg[23]_0[3:0] acc_reg[27]_0[3:0] acc_reg[31]_1[3:0] b_out_reg[4] b_out_reg[4]_0[1:0] b_out_reg[7][2:0] b_out_reg[7]_0
c_out_flat[31:0] clear_0 product__85_carry__0_i_3__14_0[1:0] product__85_carry__0_i_3__14_1[1:0] product__85_carry__0_i_4__14_0[3:0] product__85_carry__0_i_4__14_1[3:0] product__85_carry__0_i_6__14_0[3:0] product__85_carry__0_i_6__14_1[3:0] product__85_carry__1_i_2__14[1:0] product__85_carry__1_i_2__14_0[1:0] product__85_carry__1_i_3__14

product__85_carry__1_i_3__14_0

product__85_carry__1_i_5__14[3:0]

product__85_carry_i_1__14_0[3:0] product__85_carry_i_8__14_0[3:0] product__85_carry_i_8__14_1[3:0]

g_row[2].g_col[3].u_pe

pe_10

acc_reg[31]_0

b_out_reg[7]_6

clear clk product__0_carry__0_0

product__0_carry__0_1

product__0_carry__0_2
product__0_carry__0_3
product__30_carry__0_0

product__30_carry__0_1

product__30_carry__0_2
product__30_carry__0_3

CO

D[7:0] DI[2:0] O[3:0] Q[7:0] S[3:0] acc[15]_i_5__0[1:0] acc[3]_i_6__10_0[2:0] acc[3]_i_6__10_1[3:0] acc[3]_i_9__10_0[2:0] acc[15]_i_5__0_0[2:0] acc_reg[15]_0
acc_reg[15]_1[3:0] acc_reg[19]_0[3:0] acc_reg[23]_0[3:0] acc_reg[27]_0[3:0] acc_reg[31]_1[3:0] acc_reg[31]_2[19:0]

b_out_reg[0]_0
b_out_reg[0]_1

b_out_reg[1]_0[2:0] b_out_reg[1]_1[3:0] b_out_reg[1]_2[1:0] b_out_reg[2]_0[3:0] b_out_reg[3]_0

b_out_reg[3]_1

b_out_reg[4]_0

b_out_reg[4]_1[1:0] b_out_reg[4]_2[2:0] b_out_reg[4]_3[3:0] b_out_reg[4]_4[1:0] b_out_reg[5]_0[3:0] b_out_reg[6]_0

b_out_reg[7]_0[2:0] b_out_reg[7]_1
b_out_reg[7]_2
b_out_reg[7]_3

b_out_reg[7]_4[1:0] b_out_reg[7]_5[3:0] c_out_flat[31:0] clear_0[3:0] clear_1[3:0] clear_2[3:0] clear_3[3:0] clear_4[3:0] product__59_carry__0_0[1:0] product__59_carry__1_0[2:0]

product__59_carry__1_1[7:0] product__85_carry__0_i_3__10_0[1:0] product__85_carry__0_i_3__10_1[1:0] product__85_carry__0_i_4__10_0[3:0] product__85_carry__0_i_4__10_1[3:0] product__85_carry__0_i_6__10_0[3:0] product__85_carry__0_i_6__10_1[3:0] product__85_carry__1_0[1:0] product__85_carry__1_1[2:0] product__85_carry__1_2

product__85_carry__1_3

product__85_carry__1_i_2__10[1:0] product__85_carry__1_i_2__10_0[1:0] product__85_carry__1_i_3__10

product__85_carry__1_i_3__10_0 product__85_carry__1_i_5__10[3:0] product__85_carry_i_1__10_0[3:0] product__85_carry_i_8__10_0[3:0] product__85_carry_i_8__10_1[3:0]

g_row[1].g_col[3].u_pe

pe_6

acc_reg[31]_0

b_out_reg[0]_2

clear clk product__0_carry__0_0

product__0_carry__0_1

product__0_carry__0_2
product__0_carry__0_3
product__30_carry__0_0

product__30_carry__0_1

product__30_carry__0_2
product__30_carry__0_3

CO

D[7:0] DI[2:0] O[3:0] Q[7:0] S[3:0] acc[15]_i_5__2[1:0] acc[15]_i_5__2_0[2:0] acc[3]_i_6__6_0[2:0] acc[3]_i_6__6_1[3:0] acc[3]_i_9__6_0[2:0] acc_reg[15]_0[3:0] acc_reg[15]_1[3:0] acc_reg[19]_0[3:0] acc_reg[23]_0[3:0] acc_reg[27]_0[3:0] acc_reg[31]_1[3:0] acc_reg[31]_2[19:0]

b_out_reg[0]_0
b_out_reg[0]_1

b_out_reg[1]_0[2:0] b_out_reg[1]_1[3:0] b_out_reg[1]_2[1:0] b_out_reg[2]_0[3:0] b_out_reg[3]_0

b_out_reg[3]_1

b_out_reg[4]_0

b_out_reg[4]_1[1:0] b_out_reg[4]_2[2:0] b_out_reg[4]_3[3:0] b_out_reg[4]_4[1:0] b_out_reg[5]_0[3:0] b_out_reg[6]_0

b_out_reg[7]_0[2:0] b_out_reg[7]_1
b_out_reg[7]_2
b_out_reg[7]_3

b_out_reg[7]_4[1:0] b_out_reg[7]_5[3:0] c_out_flat[31:0] clear_0

clear_1[3:0] clear_2[3:0] clear_3[3:0] clear_4[3:0] product__59_carry__0_0[1:0] product__59_carry__1_0[2:0]
product__59_carry__1_1[7:0] product__85_carry__0_i_3__6_0[1:0] product__85_carry__0_i_3__6_1[1:0] product__85_carry__0_i_4__6_0[3:0] product__85_carry__0_i_4__6_1[3:0] product__85_carry__0_i_6__6_0[3:0] product__85_carry__0_i_6__6_1[3:0] product__85_carry__1_0[1:0] product__85_carry__1_1[2:0] product__85_carry__1_2

product__85_carry__1_3

product__85_carry__1_i_2__6[1:0] product__85_carry__1_i_2__6_0[1:0] product__85_carry__1_i_3__6

product__85_carry__1_i_3__6_0 product__85_carry__1_i_5__6[3:0] product__85_carry_i_1__6_0[3:0] product__85_carry_i_8__6_0[3:0] product__85_carry_i_8__6_1[3:0]

g_row[0].g_col[3].u_pe

pe_2

acc_reg[31]_0

b_out_reg[0]_2

clear clk product__0_carry__0_0

product__0_carry__0_1

product__0_carry__0_2

product__0_carry__0_3
product__30_carry__0_0
product__30_carry__0_1
product__30_carry__0_2

product__30_carry__0_3

CO

DI[2:0] O[3:0] Q[7:0] S[3:0] acc[15]_i_5__4[1:0] acc[15]_i_5__4_0[2:0] acc[3]_i_6__2_0[2:0] acc[3]_i_6__2_1[3:0] acc[3]_i_9__2_0[3:0] acc_reg[15]_0
acc_reg[15]_1[3:0] acc_reg[19]_0[3:0] acc_reg[23]_0[3:0] acc_reg[27]_0[3:0] acc_reg[31]_1[3:0] acc_reg[31]_2[19:0] b_in_flat[7:0]

b_in_flat[28] b_in_flat[28]_0[1:0] b_in_flat[31][2:0] b_in_flat[31]_0

b_out_reg[0]_0

b_out_reg[0]_1

b_out_reg[1]_0[2:0] b_out_reg[1]_1[3:0] b_out_reg[1]_2[1:0] b_out_reg[2]_0[3:0] b_out_reg[3]_0

b_out_reg[3]_1

b_out_reg[4]_0[2:0] b_out_reg[4]_1[3:0] b_out_reg[4]_2[1:0] b_out_reg[5]_0[3:0] b_out_reg[6]_0

b_out_reg[7]_0

b_out_reg[7]_1

b_out_reg[7]_2[1:0] b_out_reg[7]_3[3:0] c_out_flat[31:0] clear_0[3:0] clear_1[3:0] clear_2[3:0] clear_3[3:0] clear_4[3:0] product__59_carry__0_0[1:0] product__59_carry__1_0[2:0]

product__59_carry__1_1[7:0] product__85_carry__0_i_3__2_0[1:0] product__85_carry__0_i_3__2_1[1:0] product__85_carry__0_i_4__2_0[3:0] product__85_carry__0_i_4__2_1[3:0] product__85_carry__0_i_6__2_0[3:0] product__85_carry__0_i_6__2_1[3:0] product__85_carry__1_0[1:0] product__85_carry__1_1[2:0] product__85_carry__1_2

product__85_carry__1_3

product__85_carry__1_i_2__2[1:0] product__85_carry__1_i_2__2_0[1:0] product__85_carry__1_i_3__2

product__85_carry__1_i_3__2_0 product__85_carry__1_i_5__2[3:0] product__85_carry_i_1__2_0[2:0] product__85_carry_i_1__2_1[3:0] product__85_carry_i_8__2_0[3:0]

g_row[1].g_col[2].u_pe

pe_5

a_out_reg[0]_0 a_out_reg[2]_0

a_out_reg[2]_1

a_out_reg[3]_0

a_out_reg[3]_1
a_out_reg[4]_0
a_out_reg[4]_1

a_out_reg[5]_1

a_out_reg[5]_2
acc_reg[31]_0

clear clk product__0_carry__0_0

product__0_carry__0_1

product__0_carry__0_2
product__0_carry__0_3
product__30_carry__0_0
product__30_carry__0_1

product__30_carry__0_2

product__30_carry__0_3

CO

D[7:0] DI[2:0] O[3:0] Q[7:0] S[3:0] a_out_reg[1]_0[1:0] a_out_reg[1]_1[1:0] a_out_reg[2]_2[2:0] a_out_reg[2]_3[2:0] a_out_reg[5]_0[3:0] a_out_reg[6]_0

a_out_reg[6]_1

acc[15]_i_5__9[1:0] acc[15]_i_5__9_0[2:0] acc[3]_i_6__5_0[2:0] acc[3]_i_6__5_1[3:0] acc[3]_i_9__5_0[2:0] acc_reg[15]_0[3:0] acc_reg[15]_1[3:0] acc_reg[19]_0[3:0] acc_reg[23]_0[3:0] acc_reg[27]_0[3:0] acc_reg[31]_1[3:0] acc_reg[31]_2[19:0] b_out_reg[0]_0

b_out_reg[0]_1

b_out_reg[1]_0[2:0] b_out_reg[1]_1[3:0] b_out_reg[1]_2[1:0] b_out_reg[2]_0[3:0] b_out_reg[3]_0

b_out_reg[3]_1
b_out_reg[4]_0

b_out_reg[4]_1[1:0] b_out_reg[4]_2[2:0] b_out_reg[4]_3[3:0] b_out_reg[4]_4[1:0] b_out_reg[5]_0[3:0] b_out_reg[6]_0

b_out_reg[7]_0[2:0] b_out_reg[7]_1

b_out_reg[7]_2[7:0] b_out_reg[7]_3
b_out_reg[7]_4

b_out_reg[7]_5[1:0] b_out_reg[7]_6[3:0]

b_out_reg[7]_7[7:0]

c_out_flat[31:0] clear_0

clear_1[3:0] clear_2[3:0] clear_3[3:0] clear_4[3:0]

product__59_carry_0[7:0]

product__59_carry__0_0[1:0] product__59_carry__1_0[2:0]

product__59_carry__1_1[7:0] product__85_carry__0_i_3__5_0[1:0] product__85_carry__0_i_3__5_1[1:0] product__85_carry__0_i_4__5_0[3:0] product__85_carry__0_i_4__5_1[3:0] product__85_carry__0_i_6__5_0[3:0] product__85_carry__0_i_6__5_1[3:0] product__85_carry__1_0[1:0] product__85_carry__1_1[2:0] product__85_carry__1_2

product__85_carry__1_3

product__85_carry__1_i_2__5[1:0] product__85_carry__1_i_2__5_0[1:0] product__85_carry__1_i_3__5

product__85_carry__1_i_3__5_0

product__85_carry__1_i_5__5[3:0] product__85_carry_i_1__5_0[3:0] product__85_carry_i_8__5_0[3:0] product__85_carry_i_8__5_1[3:0]

g_row[2].g_col[1].u_pe

pe_8

a_out_reg[2]_0

a_out_reg[2]_1

a_out_reg[3]_0

a_out_reg[3]_1
a_out_reg[4]_0
a_out_reg[4]_1

a_out_reg[5]_1

a_out_reg[5]_2 acc_reg[31]_0

b_out_reg[0]_2

clear clk product__0_carry__0_0

product__0_carry__0_1

product__0_carry__0_2
product__0_carry__0_3
product__30_carry__0_0
product__30_carry__0_1

product__30_carry__0_2

product__30_carry__0_3

CO D[7:0] DI[2:0] O[3:0] Q[7:0] S[3:0] a_out_reg[1]_0[1:0] a_out_reg[1]_1[1:0] a_out_reg[2]_2[2:0] a_out_reg[2]_3[2:0] a_out_reg[5]_0[3:0] a_out_reg[6]_0

a_out_reg[6]_1

acc[15]_i_5__1[1:0] acc[15]_i_5__1_0[2:0] acc[3]_i_6__8_0[2:0] acc[3]_i_6__8_1[3:0] acc[3]_i_9__8_0[2:0] acc_reg[15]_0

acc_reg[15]_1[3:0] acc_reg[19]_0[3:0] acc_reg[23]_0[3:0] acc_reg[27]_0[3:0] acc_reg[31]_1[3:0] acc_reg[31]_2[19:0] b_out_reg[0]_0

b_out_reg[0]_1

b_out_reg[1]_0[2:0] b_out_reg[1]_1[3:0] b_out_reg[1]_2[1:0] b_out_reg[2]_0[3:0] b_out_reg[3]_0

b_out_reg[3]_1
b_out_reg[4]_0

b_out_reg[4]_1[1:0] b_out_reg[4]_2[2:0] b_out_reg[4]_3[3:0] b_out_reg[4]_4[1:0] b_out_reg[5]_0[3:0] b_out_reg[6]_0

b_out_reg[7]_0[2:0] b_out_reg[7]_1

b_out_reg[7]_2[7:0] b_out_reg[7]_3
b_out_reg[7]_4

b_out_reg[7]_5[1:0] b_out_reg[7]_6[3:0]

b_out_reg[7]_7[7:0]

c_out_flat[31:0] clear_0[3:0] clear_1[3:0] clear_2[3:0] clear_3[3:0] clear_4[3:0]

product__59_carry_0[7:0]

product__59_carry__0_0[1:0] product__59_carry__1_0[2:0]

product__59_carry__1_1[7:0] product__85_carry__0_i_3__8_0[1:0] product__85_carry__0_i_3__8_1[1:0] product__85_carry__0_i_4__8_0[3:0] product__85_carry__0_i_4__8_1[3:0] product__85_carry__0_i_6__8_0[3:0] product__85_carry__0_i_6__8_1[3:0] product__85_carry__1_0[1:0] product__85_carry__1_1[2:0] product__85_carry__1_2

product__85_carry__1_3

product__85_carry__1_i_2__8[1:0] product__85_carry__1_i_2__8_0[1:0] product__85_carry__1_i_3__8

product__85_carry__1_i_3__8_0

product__85_carry__1_i_5__8[3:0] product__85_carry_i_1__8_0[3:0] product__85_carry_i_8__8_0[3:0] product__85_carry_i_8__8_1[3:0]

g_row[1].g_col[1].u_pe

pe_4

a_out_reg[0]_0 a_out_reg[2]_0
a_out_reg[2]_1

a_out_reg[3]_0
a_out_reg[3]_1
a_out_reg[4]_0

a_out_reg[4]_1

a_out_reg[5]_1
a_out_reg[5]_2

acc_reg[31]_0

clear clk product__0_carry__0_0

product__0_carry__0_1
product__0_carry__0_2
product__0_carry__0_3

product__30_carry__0_0

product__30_carry__0_1
product__30_carry__0_2
product__30_carry__0_3

CO

D[7:0] DI[2:0] O[3:0] Q[7:0] S[3:0] a_out_reg[1]_0[1:0] a_out_reg[1]_1[1:0] a_out_reg[2]_2[2:0] a_out_reg[2]_3[2:0] a_out_reg[5]_0[3:0] a_out_reg[6]_0

a_out_reg[6]_1

acc[15]_i_5__3[1:0] acc[15]_i_5__3_0[2:0] acc[3]_i_6__4_0[2:0] acc[3]_i_6__4_1[3:0] acc[3]_i_9__4_0[2:0] acc_reg[15]_0[3:0] acc_reg[15]_1[3:0] acc_reg[19]_0[3:0] acc_reg[23]_0[3:0] acc_reg[27]_0[3:0] acc_reg[31]_1[3:0] acc_reg[31]_2[19:0] b_out_reg[0]_0

b_out_reg[0]_1

b_out_reg[1]_0[2:0] b_out_reg[1]_1[3:0] b_out_reg[1]_2[1:0] b_out_reg[2]_0[3:0] b_out_reg[3]_0

b_out_reg[3]_1

b_out_reg[4]_0

b_out_reg[4]_1[1:0] b_out_reg[4]_2[2:0] b_out_reg[4]_3[3:0] b_out_reg[4]_4[1:0] b_out_reg[5]_0[3:0] b_out_reg[6]_0

b_out_reg[7]_0[2:0] b_out_reg[7]_1

b_out_reg[7]_2[7:0] b_out_reg[7]_3
b_out_reg[7]_4

b_out_reg[7]_5[1:0] b_out_reg[7]_6[3:0]

b_out_reg[7]_7[7:0]

c_out_flat[31:0] clear_0

clear_1[3:0] clear_2[3:0] clear_3[3:0] clear_4[3:0]

product__59_carry_0[7:0]

product__59_carry__0_0[1:0] product__59_carry__1_0[2:0]

product__59_carry__1_1[7:0] product__85_carry__0_i_3__4_0[1:0] product__85_carry__0_i_3__4_1[1:0] product__85_carry__0_i_4__4_0[3:0] product__85_carry__0_i_4__4_1[3:0] product__85_carry__0_i_6__4_0[3:0] product__85_carry__0_i_6__4_1[3:0] product__85_carry__1_0[1:0] product__85_carry__1_1[2:0] product__85_carry__1_2

product__85_carry__1_3

product__85_carry__1_i_2__4[1:0] product__85_carry__1_i_2__4_0[1:0] product__85_carry__1_i_3__4

product__85_carry__1_i_3__4_0

product__85_carry__1_i_5__4[3:0] product__85_carry_i_1__4_0[3:0] product__85_carry_i_8__4_0[3:0] product__85_carry_i_8__4_1[3:0]

g_row[0].g_col[2].u_pe

pe_1

acc_reg[31]_0

b_out_reg[0]_2

clear clk product__0_carry__0_0

product__0_carry__0_1
product__0_carry__0_2
product__0_carry__0_3

product__30_carry__0_0

product__30_carry__0_1
product__30_carry__0_2
product__30_carry__0_3

CO

D[7:0] DI[2:0] O[3:0] Q[7:0] S[3:0] a_out_reg[1]_0[2:0] a_out_reg[1]_1[3:0] a_out_reg[2]_0[3:0] a_out_reg[2]_1[3:0] a_out_reg[5]_0[3:0] a_out_reg[6]_0[1:0] a_out_reg[6]_1[1:0] acc[15]_i_5__7[1:0] acc[15]_i_5__7_0[2:0] acc[3]_i_6__1_0[2:0] acc[3]_i_6__1_1[3:0] acc[3]_i_9__1_0[2:0] acc[3]_i_9__1_1[3:0] acc_reg[15]_0[3:0] acc_reg[15]_1
acc_reg[15]_2[3:0] acc_reg[15]_3[3:0] acc_reg[19]_0[3:0] acc_reg[23]_0[3:0] acc_reg[27]_0[3:0] acc_reg[31]_1[3:0] acc_reg[31]_2[39:0] b_in_flat[15:0]

b_in_flat[20] b_in_flat[20]_0[1:0] b_in_flat[23][2:0] b_in_flat[23]_0

b_in_flat[25][1:0] b_in_flat[26][3:0] b_in_flat[28][2:0] b_in_flat[28]_0[3:0] b_in_flat[28]_1[1:0] b_in_flat[29][3:0] b_in_flat[30] b_in_flat[31] b_in_flat[31]_0[3:0] b_out_reg[0]_0

b_out_reg[0]_1

b_out_reg[1]_0[2:0] b_out_reg[1]_1[3:0] b_out_reg[1]_2[1:0] b_out_reg[2]_0[3:0] b_out_reg[3]_0

b_out_reg[3]_1

b_out_reg[4]_0[2:0] b_out_reg[4]_1[3:0] b_out_reg[4]_2[1:0] b_out_reg[5]_0[3:0] b_out_reg[6]_0

b_out_reg[7]_0

b_out_reg[7]_1

b_out_reg[7]_2[1:0] b_out_reg[7]_3[3:0] c_out_flat[31:0] clear_0

clear_1[3:0] clear_2[3:0] clear_3[3:0] clear_4[3:0] clear_5[3:0] clear_6[3:0] clear_7[3:0] clear_8[3:0] clear_9[3:0] product__59_carry__0_0[1:0] product__59_carry__0_1[1:0] product__59_carry__1_0[2:0] product__59_carry__1_1[2:0]

product__59_carry__1_2[7:0] product__85_carry__0_i_3__1_0[1:0] product__85_carry__0_i_3__1_1[1:0] product__85_carry__0_i_4__1_0[3:0] product__85_carry__0_i_4__1_1[3:0] product__85_carry__0_i_6__1_0[3:0] product__85_carry__0_i_6__1_1[3:0] product__85_carry__1_0[1:0] product__85_carry__1_1[2:0] product__85_carry__1_2[1:0] product__85_carry__1_3[2:0] product__85_carry__1_4

product__85_carry__1_5
product__85_carry__1_6
product__85_carry__1_7

product__85_carry__1_i_2__1[1:0] product__85_carry__1_i_2__1_0[1:0] product__85_carry__1_i_3__1

product__85_carry__1_i_3__1_0

product__85_carry__1_i_5__1[3:0] product__85_carry_i_1__1_0[2:0] product__85_carry_i_1__1_1[3:0] product__85_carry_i_8__1_0[3:0] product__85_carry_i_8__1_1[3:0]

g_row[0].g_col[1].u_pe

pe_0

acc_reg[31]_0

b_out_reg[0]_2

clear clk product__0_carry__0_0

product__0_carry__0_1
product__0_carry__0_2
product__0_carry__0_3

product__30_carry__0_0
product__30_carry__0_1

product__30_carry__0_2
product__30_carry__0_3

CO

D[7:0] DI[2:0] O[3:0] Q[7:0] S[3:0] a_out_reg[1]_0[2:0] a_out_reg[1]_1[3:0] a_out_reg[2]_0[3:0] a_out_reg[2]_1[3:0] a_out_reg[5]_0[3:0] a_out_reg[6]_0[1:0] a_out_reg[6]_1[1:0] acc[15]_i_5__5[1:0] acc[15]_i_5__5_0[2:0] acc[3]_i_6__0_0[2:0] acc[3]_i_6__0_1[3:0] acc[3]_i_9__0_0[3:0] acc_reg[15]_0

acc_reg[15]_1

acc_reg[15]_2[3:0] acc_reg[15]_3[3:0] acc_reg[19]_0[3:0] acc_reg[23]_0[3:0] acc_reg[27]_0[3:0] acc_reg[31]_1[3:0] acc_reg[31]_2[39:0] b_in_flat[15:0]

b_in_flat[12] b_in_flat[12]_0[1:0] b_in_flat[15][2:0] b_in_flat[15]_0

b_in_flat[17][2:0] b_in_flat[17]_0[3:0] b_in_flat[17]_1[1:0] b_in_flat[18][3:0] b_in_flat[20][2:0] b_in_flat[20]_0[3:0] b_in_flat[20]_1[1:0] b_in_flat[21][3:0] b_in_flat[22] b_in_flat[23] b_in_flat[23]_0[3:0] b_out_reg[0]_0

b_out_reg[0]_1

b_out_reg[1]_0[2:0] b_out_reg[1]_1[3:0] b_out_reg[1]_2[1:0] b_out_reg[2]_0[3:0] b_out_reg[3]_0

b_out_reg[3]_1

b_out_reg[4]_0[2:0] b_out_reg[4]_1[3:0] b_out_reg[4]_2[1:0] b_out_reg[5]_0[3:0] b_out_reg[6]_0

b_out_reg[7]_0[7:0] b_out_reg[7]_1
b_out_reg[7]_2

b_out_reg[7]_3[1:0] b_out_reg[7]_4[3:0] c_out_flat[31:0] clear_0[3:0] clear_1[3:0] clear_2[3:0] clear_3[3:0] clear_4[3:0] clear_5[3:0] clear_6[3:0] clear_7[3:0] clear_8[3:0] clear_9[3:0] product__59_carry__0_0[1:0] product__59_carry__0_1[1:0] product__59_carry__1_0[2:0] product__59_carry__1_1[2:0]

product__59_carry__1_2[7:0] product__85_carry__0_i_3__0_0[1:0] product__85_carry__0_i_3__0_1[1:0] product__85_carry__0_i_4__0_0[3:0] product__85_carry__0_i_4__0_1[3:0] product__85_carry__0_i_6__0_0[3:0] product__85_carry__0_i_6__0_1[3:0] product__85_carry__1_0[1:0] product__85_carry__1_1[2:0] product__85_carry__1_2[1:0] product__85_carry__1_3[2:0] product__85_carry__1_4

product__85_carry__1_5
product__85_carry__1_6
product__85_carry__1_7

product__85_carry__1_i_2__0[1:0] product__85_carry__1_i_2__0_0[1:0] product__85_carry__1_i_3__0

product__85_carry__1_i_3__0_0

product__85_carry__1_i_5__0[3:0] product__85_carry_i_1__0_0[2:0] product__85_carry_i_1__0_1[3:0] product__85_carry_i_8__0_0[3:0]

g_row[1].g_col[0].u_pe

pe_3

a_out_reg[2]_0

a_out_reg[2]_1

a_out_reg[3]_0

a_out_reg[3]_1
a_out_reg[4]_0

a_out_reg[4]_1

a_out_reg[5]_1

a_out_reg[5]_2

acc_reg[31]_0

b_out_reg[0]_0
clear

CO

clk

D[7:0] DI[2:0] O[3:0] Q[7:0] S

a_in_flat[15:0] a_in_flat[17][2:0] a_in_flat[17]_0[3:0] a_in_flat[18][3:0] a_in_flat[18]_0[3:0] a_in_flat[21][3:0] a_in_flat[22][1:0] a_in_flat[22]_0[1:0] a_out_reg[1]_0[1:0] a_out_reg[1]_1[1:0] a_out_reg[2]_2[2:0] a_out_reg[2]_3[2:0] a_out_reg[5]_0[3:0] a_out_reg[6]_0

acc[15]_i_5__8[1:0] acc[15]_i_5__8_0[2:0] acc[3]_i_6__3_0[2:0] acc[3]_i_6__3_1[3:0] acc[3]_i_9__3_0[2:0] acc[3]_i_9__3_1[3:0] acc_reg[15]_0[3:0] acc_reg[15]_1[3:0] acc_reg[19]_0[3:0] acc_reg[23]_0[3:0] acc_reg[27]_0[3:0] acc_reg[31]_1[3:0] acc_reg[31]_2[19:0] b_out_reg[1]_0[3:0] b_out_reg[1]_1[1:0] b_out_reg[2]_0[3:0] b_out_reg[4]_0

b_out_reg[4]_1[1:0] b_out_reg[4]_2[2:0] b_out_reg[4]_3[3:0] b_out_reg[4]_4[1:0] b_out_reg[5]_0[3:0] b_out_reg[6]_0

b_out_reg[7]_0[2:0] b_out_reg[7]_1

b_out_reg[7]_2[7:0] b_out_reg[7]_3

b_out_reg[7]_4[3:0] c_out_flat[31:0] clear_0

clear_1[3:0] clear_2[3:0] clear_3[3:0] clear_4[3:0]

product__59_carry_0[7:0]

product__59_carry__0_0[1:0] product__59_carry__1_0[2:0]

product__85_carry__0_i_3__3_0[1:0] product__85_carry__0_i_3__3_1[1:0] product__85_carry__0_i_4__3_0[3:0] product__85_carry__0_i_4__3_1[3:0] product__85_carry__0_i_6__3_0[3:0] product__85_carry__0_i_6__3_1[3:0] product__85_carry__1_0[1:0] product__85_carry__1_1[2:0] product__85_carry__1_2

product__85_carry__1_3

product__85_carry__1_i_2__3[1:0] product__85_carry__1_i_2__3_0[1:0] product__85_carry__1_i_3__3

product__85_carry__1_i_3__3_0

product__85_carry__1_i_5__3[3:0]

product__85_carry_i_1__3_0[2:0] product__85_carry_i_1__3_1[3:0] product__85_carry_i_8__3_0[3:0] product__85_carry_i_8__3_1[3:0]

g_row[2].g_col[0].u_pe

pe_7

a_out_reg[0]_0

a_out_reg[2]_0

a_out_reg[2]_1

a_out_reg[3]_0

a_out_reg[3]_1

a_out_reg[4]_0
a_out_reg[4]_1

a_out_reg[5]_1

a_out_reg[5]_2

acc_reg[31]_0

clear clk

CO

D[7:0] DI[2:0] O[3:0] Q[7:0] S

a_in_flat[15:0] a_in_flat[25][2:0] a_in_flat[25]_0[3:0] a_in_flat[26][3:0] a_in_flat[26]_0[3:0] a_in_flat[29][3:0] a_in_flat[30][1:0] a_in_flat[30]_0[1:0] a_out_reg[1]_0[1:0] a_out_reg[1]_1[1:0] a_out_reg[2]_2[2:0] a_out_reg[2]_3[2:0] a_out_reg[5]_0[3:0] a_out_reg[6]_0

acc[15]_i_5__10[1:0] acc[15]_i_5__10_0[2:0] acc[3]_i_6__7_0[2:0] acc[3]_i_6__7_1[3:0] acc[3]_i_9__7_0[3:0] acc_reg[15]_0

acc_reg[15]_1[3:0] acc_reg[19]_0[3:0] acc_reg[23]_0[3:0] acc_reg[27]_0[3:0] acc_reg[31]_1[3:0] acc_reg[31]_2[19:0] b_out_reg[1]_0[2:0] b_out_reg[1]_1[3:0] b_out_reg[1]_2[1:0] b_out_reg[2]_0[3:0] b_out_reg[4]_0

b_out_reg[4]_1[1:0] b_out_reg[4]_2[2:0] b_out_reg[4]_3[3:0] b_out_reg[4]_4[1:0] b_out_reg[5]_0[3:0] b_out_reg[6]_0

b_out_reg[7]_0[2:0] b_out_reg[7]_1

b_out_reg[7]_2

b_out_reg[7]_3[3:0] c_out_flat[31:0] clear_0[3:0] clear_1[3:0] clear_2[3:0] clear_3[3:0] clear_4[3:0]

product__59_carry_0[7:0]

product__59_carry__0_0[1:0] product__59_carry__1_0[2:0]

product__85_carry__0_i_3__7_0[1:0] product__85_carry__0_i_3__7_1[1:0] product__85_carry__0_i_4__7_0[3:0] product__85_carry__0_i_4__7_1[3:0] product__85_carry__0_i_6__7_0[3:0] product__85_carry__0_i_6__7_1[3:0] product__85_carry__1_0[1:0] product__85_carry__1_1[2:0] product__85_carry__1_2

product__85_carry__1_3

product__85_carry__1_i_2__7[1:0] product__85_carry__1_i_2__7_0[1:0] product__85_carry__1_i_3__7

product__85_carry__1_i_3__7_0

product__85_carry__1_i_5__7[3:0]

product__85_carry_i_1__7_0[2:0] product__85_carry_i_1__7_1[3:0] product__85_carry_i_8__7_0[3:0] product__85_carry_i_8__7_1[3:0]

g_row[3].g_col[0].u_pe

pe_11

a_out_reg[0]_0

a_out_reg[2]_0

a_out_reg[2]_1

a_out_reg[3]_0

a_out_reg[3]_1

a_out_reg[4]_0
a_out_reg[4]_1

a_out_reg[5]_1

a_out_reg[5]_2

acc_reg[31]_0

clear clk

DI[1:0] Q[7:0] S

a_in_flat[7:0] a_out_reg[1]_0[1:0] a_out_reg[2]_2[2:0] a_out_reg[2]_3[2:0] a_out_reg[5]_0[3:0] a_out_reg[6]_0

acc[15]_i_5__12[1:0] acc[15]_i_5__12_0[2:0] acc[3]_i_6__11_0[2:0] acc[3]_i_6__11_1[3:0] acc[3]_i_9__11_0[2:0] acc[3]_i_9__11_1[3:0] acc_reg[15]_0[3:0] acc_reg[19]_0[3:0] acc_reg[23]_0[3:0] acc_reg[27]_0[3:0] acc_reg[31]_1[3:0] b_out_reg[4] b_out_reg[4]_0[1:0] b_out_reg[7][2:0] b_out_reg[7]_0

c_out_flat[31:0] clear_0

product__59_carry_0[7:0] product__85_carry__0_i_3__11_0[1:0] product__85_carry__0_i_3__11_1[1:0] product__85_carry__0_i_4__11_0[3:0] product__85_carry__0_i_4__11_1[3:0] product__85_carry__0_i_6__11_0[3:0] product__85_carry__0_i_6__11_1[3:0] product__85_carry__1_i_2__11[1:0] product__85_carry__1_i_2__11_0[1:0] product__85_carry__1_i_3__11
product__85_carry__1_i_3__11_0 product__85_carry__1_i_5__11[3:0] product__85_carry_i_1__11_0[2:0] product__85_carry_i_1__11_1[3:0] product__85_carry_i_8__11_0[3:0] product__85_carry_i_8__11_1[3:0]

g_row[3].g_col[1].u_pe

pe_12

a_out_reg[0]_0

a_out_reg[2]_0

a_out_reg[2]_1

a_out_reg[3]_0

a_out_reg[3]_1

a_out_reg[4]_0
a_out_reg[4]_1

a_out_reg[5]_1

a_out_reg[5]_2

clear

clear_0

clken
D[7:0] DI[2:0] Q[7:0] S[3:0] a_out_reg[1]_0[1:0] a_out_reg[1]_1[1:0] a_out_reg[2]_2[2:0] a_out_reg[2]_3[2:0] a_out_reg[5]_0[3:0] a_out_reg[6]_0

a_out_reg[6]_1

acc[15]_i_5[1:0] acc[15]_i_5_0[2:0] acc[3]_i_6__12_0[2:0] acc[3]_i_6__12_1[3:0] acc[3]_i_9__12_0[2:0] acc_reg[15]_0[3:0] acc_reg[19]_0[3:0] acc_reg[23]_0[3:0] acc_reg[27]_0[3:0] acc_reg[31]_0[3:0] b_out_reg[4] b_out_reg[4]_0[1:0] b_out_reg[7][2:0] b_out_reg[7]_0

c_out_flat[31:0] clear_1

product__59_carry_0[7:0] product__85_carry__0_i_3__12_0[1:0] product__85_carry__0_i_3__12_1[1:0] product__85_carry__0_i_4__12_0[3:0] product__85_carry__0_i_4__12_1[3:0] product__85_carry__0_i_6__12_0[3:0] product__85_carry__0_i_6__12_1[3:0] product__85_carry__1_i_2__12[1:0] product__85_carry__1_i_2__12_0[1:0] product__85_carry__1_i_3__12

product__85_carry__1_i_3__12_0

product__85_carry__1_i_5__12[3:0] product__85_carry_i_1__12_0[3:0] product__85_carry_i_8__12_0[3:0] product__85_carry_i_8__12_1[3:0]

g_row[2].g_col[2].u_pe

pe_9

a_out_reg[2]_0

a_out_reg[2]_1

a_out_reg[3]_0

a_out_reg[3]_1
a_out_reg[4]_0
a_out_reg[4]_1

a_out_reg[5]_1

a_out_reg[5]_2 acc_reg[31]_0

clear clk product__0_carry__0_0

product__0_carry__0_1

product__0_carry__0_2

product__0_carry__0_3
product__30_carry__0_0
product__30_carry__0_1
product__30_carry__0_2

product__30_carry__0_3

rst_n

CO

rst_n_0

D[7:0] DI[2:0] O[3:0] Q[7:0] S[3:0] a_out_reg[1]_0[1:0] a_out_reg[1]_1[1:0] a_out_reg[2]_2[2:0] a_out_reg[2]_3[2:0] a_out_reg[5]_0[3:0] a_out_reg[6]_0

a_out_reg[6]_1

acc[15]_i_5__11[1:0] acc[15]_i_5__11_0[2:0] acc[3]_i_6__9_0[2:0] acc[3]_i_6__9_1[3:0] acc[3]_i_9__9_0[2:0] acc_reg[15]_0

acc_reg[15]_1[3:0] acc_reg[19]_0[3:0] acc_reg[23]_0[3:0] acc_reg[27]_0[3:0] acc_reg[31]_1[3:0] acc_reg[31]_2[19:0] b_out_reg[0]_0

b_out_reg[0]_1

b_out_reg[1]_0[2:0] b_out_reg[1]_1[3:0] b_out_reg[1]_2[1:0] b_out_reg[2]_0[3:0] b_out_reg[3]_0

b_out_reg[3]_1
b_out_reg[4]_0

b_out_reg[4]_1[1:0] b_out_reg[4]_2[2:0] b_out_reg[4]_3[3:0] b_out_reg[4]_4[1:0] b_out_reg[5]_0[3:0] b_out_reg[6]_0

b_out_reg[7]_0[2:0] b_out_reg[7]_1

b_out_reg[7]_2[7:0] b_out_reg[7]_3
b_out_reg[7]_4

b_out_reg[7]_5[1:0] b_out_reg[7]_6[3:0]

b_out_reg[7]_7[7:0]

c_out_flat[31:0] clear_0[3:0] clear_1[3:0] clear_2[3:0] clear_3[3:0] clear_4[3:0]

product__59_carry_0[7:0]

product__59_carry__0_0[1:0] product__59_carry__1_0[2:0]

product__59_carry__1_1[7:0] product__85_carry__0_i_3__9_0[1:0] product__85_carry__0_i_3__9_1[1:0] product__85_carry__0_i_4__9_0[3:0] product__85_carry__0_i_4__9_1[3:0] product__85_carry__0_i_6__9_0[3:0] product__85_carry__0_i_6__9_1[3:0] product__85_carry__1_0[1:0] product__85_carry__1_1[2:0] product__85_carry__1_2

product__85_carry__1_3

product__85_carry__1_i_2__9[1:0] product__85_carry__1_i_2__9_0[1:0] product__85_carry__1_i_3__9

product__85_carry__1_i_3__9_0

product__85_carry__1_i_5__9[3:0] product__85_carry_i_1__9_0[3:0] product__85_carry_i_8__9_0[3:0] product__85_carry_i_8__9_1[3:0]

g_row[0].g_col[0].u_pe

pe

a_out_reg[0]_0

acc_reg[31]_0

clear

CO

clk

DI[2:0] O[3:0] Q[7:0] S[3:0]

a_in_flat[15:0]

a_in_flat[9][2:0] a_in_flat[9]_0[3:0] a_in_flat[10][3:0] a_in_flat[10]_0[3:0] a_in_flat[13][3:0] a_in_flat[14][1:0] a_in_flat[14]_0[1:0] a_out_reg[1]_0[2:0] a_out_reg[1]_1[3:0] a_out_reg[2]_0[3:0] a_out_reg[2]_1[3:0] a_out_reg[5]_0[3:0] a_out_reg[6]_0[1:0] a_out_reg[6]_1[1:0] acc_reg[15]_0

acc_reg[15]_1[3:0] acc_reg[15]_2[3:0] acc_reg[31]_1[39:0] b_in_flat[15:0]

b_in_flat[9][1:0] b_in_flat[10][3:0] b_in_flat[12][2:0] b_in_flat[12]_0[3:0] b_in_flat[12]_1[1:0] b_in_flat[13][3:0] b_in_flat[14] b_in_flat[15] b_in_flat[15]_0[3:0] b_out_reg[1]_0[2:0] b_out_reg[1]_1[3:0] b_out_reg[1]_2[1:0] b_out_reg[2]_0[3:0] b_out_reg[4]_0[2:0] b_out_reg[4]_1[3:0] b_out_reg[4]_2[1:0] b_out_reg[5]_0[3:0] b_out_reg[6]_0

b_out_reg[7]_0[7:0] b_out_reg[7]_1

b_out_reg[7]_2[3:0] c_out_flat[31:0] clear_0[3:0] clear_1[3:0] clear_2[3:0] clear_3[3:0] clear_4[3:0] clear_5[3:0] clear_6[3:0] clear_7[3:0] clear_8[3:0] product__59_carry__0_0[1:0] product__59_carry__0_1[1:0] product__59_carry__1_0[2:0] product__59_carry__1_1[2:0]

product__85_carry__1_0[1:0] product__85_carry__1_1[2:0] product__85_carry__1_2[1:0] product__85_carry__1_3[2:0] product__85_carry__1_4

product__85_carry__1_5
product__85_carry__1_6
product__85_carry__1_7

c_out_flat[511:0]